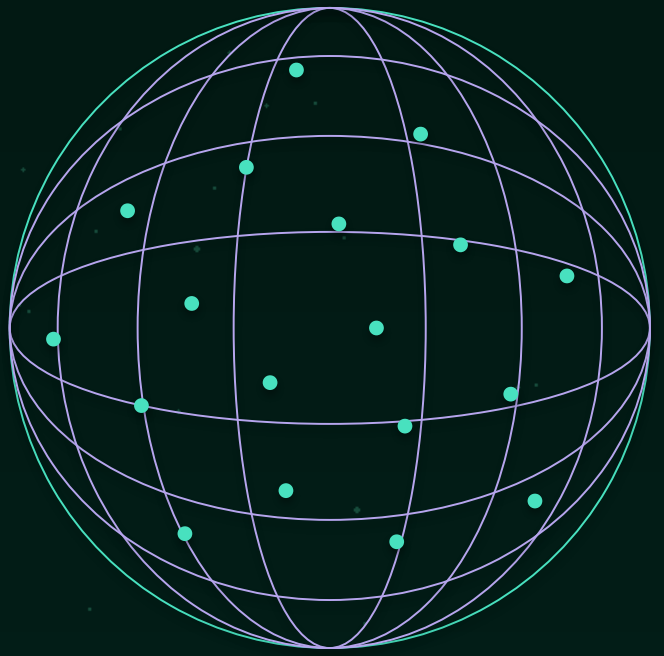
图溯

面向数字图像可信传播的来源认证与溯源方案

生成式水印 × 匿名登记 × 传播后核验

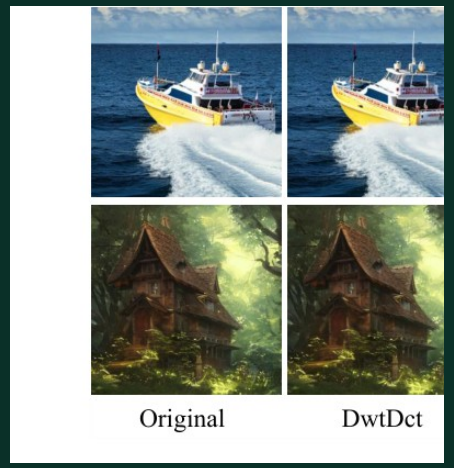


球形水印 · ICLR 2026

从内容生成，到传播后的可信核验

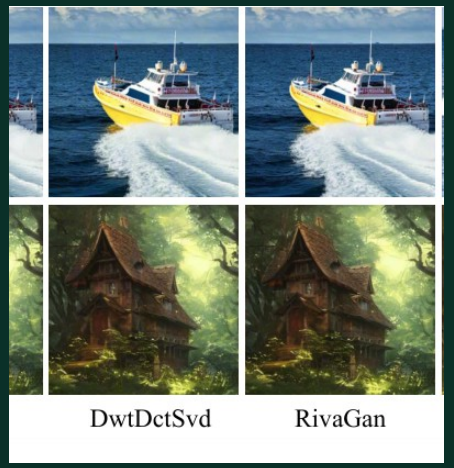
项目背景

图像创造价值，可信身份却未必随传播保留



创意设计

作品流转与署名



媒体传播

来源线索与核验



数字文创

复制使用与授权



商业营销

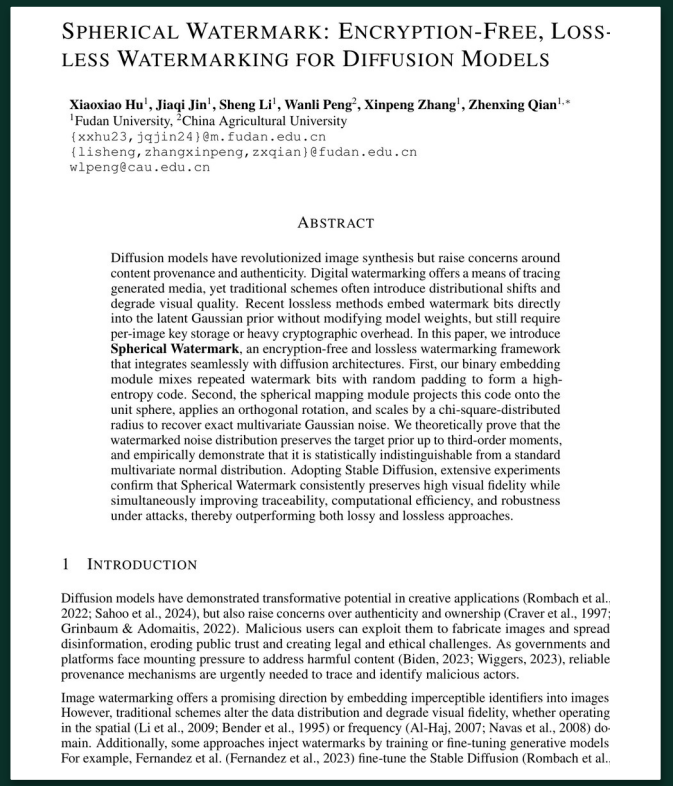
交付版本与品牌

图片被保存、转发和二次编辑之后，文件名、附带说明与来源声明可能分离。

核心问题：图像内容易复制，来源身份难持续核验

研究背景

生成能力跃升，放大了来源认证与知识产权核验的需求



SPHERICAL WATERMARK: ENCRYPTION-FREE, LOSS-LESS WATERMARKING FOR DIFFUSION MODELS

Xiaoxiao Hu[1], **Jiaqi Jin**[1], **Sheng Li**[1], **Wanli Peng**[2], **Xinpeng Zhang**[1], **Zhenxing Qian**[1,*]
[1]Fudan University, [2]China Agricultural University
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wlpeng@cau.edu.cn

ABSTRACT

Diffusion models have revolutionized image synthesis but raise concerns around content provenance and authenticity. Digital watermarking offers a means of tracing generated media, yet traditional schemes often introduce distributional shifts and degrade visual quality. Recent lossless methods embed watermark bits directly into the latent Gaussian prior without modifying model weights, but still require per-image key storage or heavy cryptographic overhead. In this paper, we introduce **Spherical Watermark**, an encryption-free and lossless watermarking framework that integrates seamlessly with diffusion architectures. First, our binary embedding module mixes repeated watermark bits with random padding to form a high-entropy code. Second, the spherical mapping module projects this code onto the unit sphere, applies an orthogonal rotation, and scales by a chi-square-distributed radius to recover exact multivariate Gaussian noise. We theoretically prove that the watermarked noise distribution preserves the target prior up to third-order moments, and empirically demonstrate that it is statistically indistinguishable from a standard multivariate normal distribution. Adopting Stable Diffusion, extensive experiments confirm that Spherical Watermark consistently preserves high visual fidelity while simultaneously improving traceability, computational efficiency, and robustness under attacks, thereby outperforming both lossy and lossless approaches.

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Image watermarking offers a promising direction by embedding imperceptible identifiers into images However, traditional schemes alter the data distribution and degrade visual fidelity, whether operating in the spatial (Li et al., 2009; Bender et al., 1995) or frequency (Al-Haj, 2007; Navas et al., 2008) domain. Additionally, some approaches inject watermarks by training or fine-tuning generative models For example, Fernandez et al. (Fernandez et al., 2023) fine-tune the Stable Diffusion (Rombach et al.,

01 高质量生成走向普及

扩散模型降低图像创作门槛，逼真内容更易生产。

02 人工识别难承担认证职责

仅凭视觉外观，难确定图片使用了哪个生成系统。

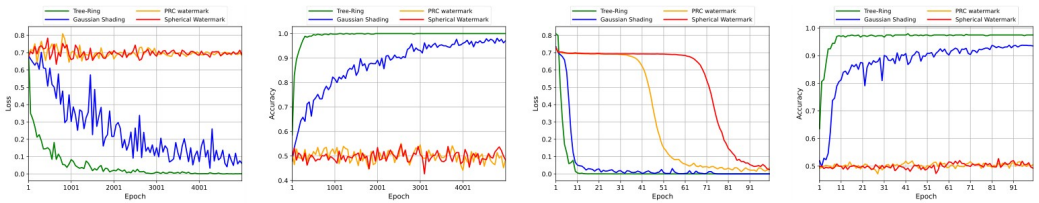
03 主动嵌入可核验身份

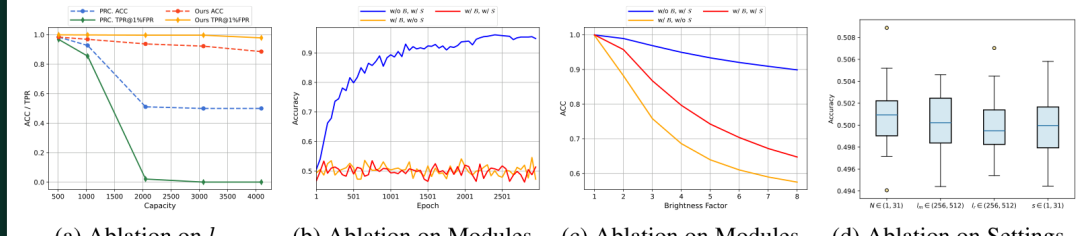
在生成时留下隐形信号，为后续归因提供线索。

认证回答“标识是否匹配”，溯源回答“关联哪条来源记录”

场景需求

可信传播需要同时满足视觉质量与标识恢复能力





(a) Ablation on l_m. (b) Ablation on Modules. (c) Ablation on Modules. (d) Ablation on Settings.

视觉侧：减少可感知与统计扰动

设计与发布：保留画面质量

审核与使用：获得可靠来源线索

核验侧：评估反演误差与消息恢复

任务调研：访谈与使用观察

评价目标：来源查找与使用决策

把“看起来不变”和“传播后可查”放在同一套验证体系中

关键痛点

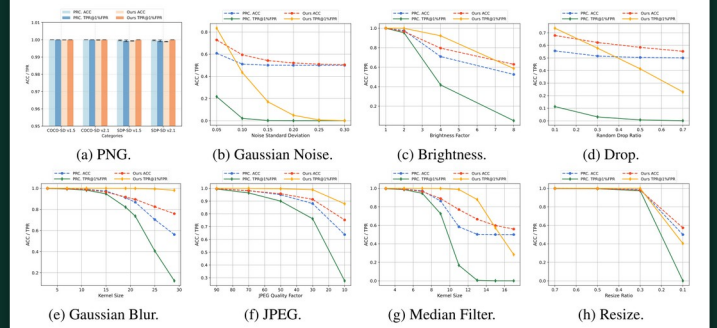
信息会分离，图像会变换，水印也有失败边界

画质扰动



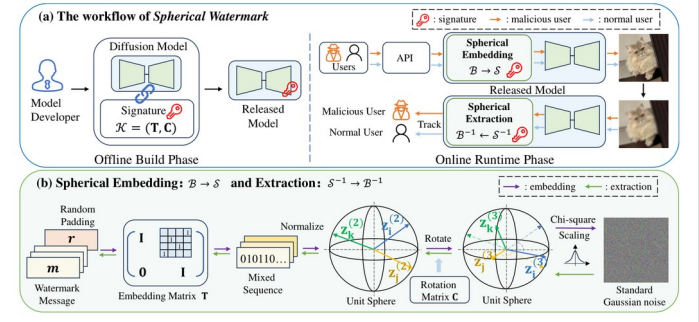
不应为身份标识明显牺牲图像观感

传播变换



压缩、缩放与裁剪可能损伤恢复能力

身份与权利



检出标识后仍需核对登记与授权材料

需要“算法保护＋登记关联＋人工复核”的完整闭环

解决思路

生成端嵌入身份，接收端直接上传核验

01 生成时建立身份

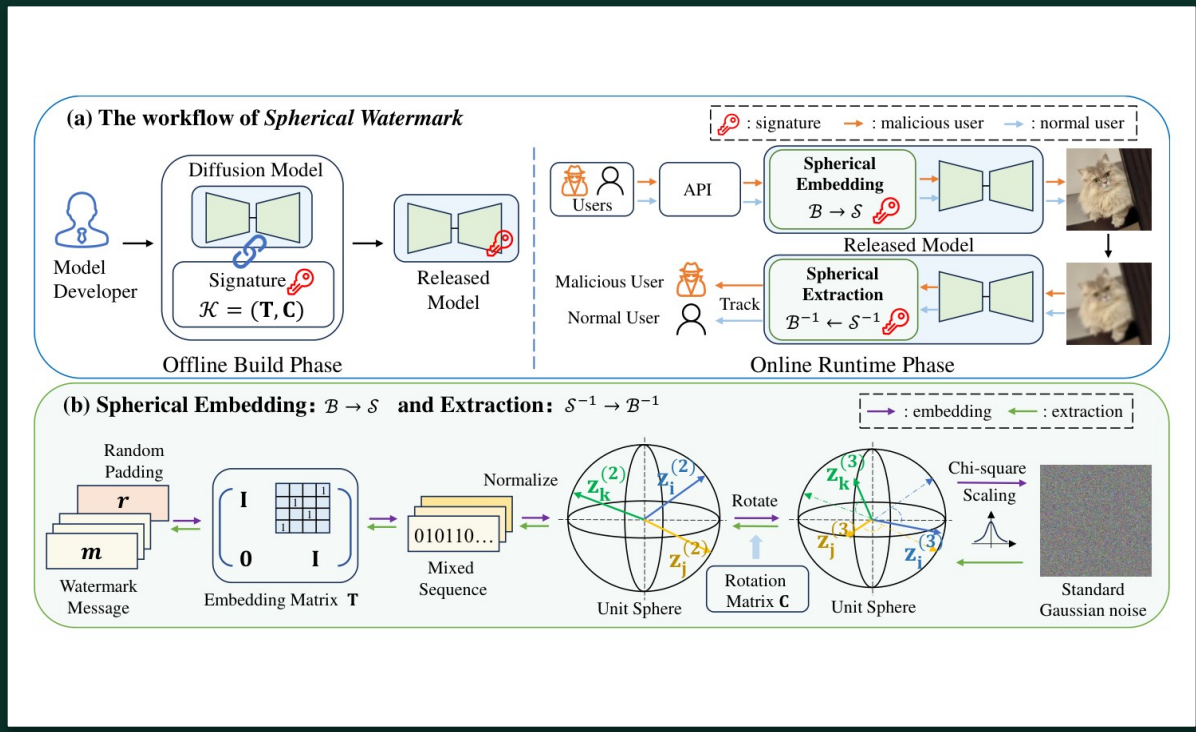
作品编号映射为噪声，再输入扩散模型。

02 传播后恢复标识

上传收到的图片，反演并提取匿名编号。

03 关联声明与人工复核

显示登记线索；不确定结果进入人工审核。



无需用户预先说明图片经历过哪些传播操作

核心方法一

重复编码与可逆混合：为身份恢复提供冗余

1

消息准备

匿名编号 m
重复 N 次＋随机填充

2

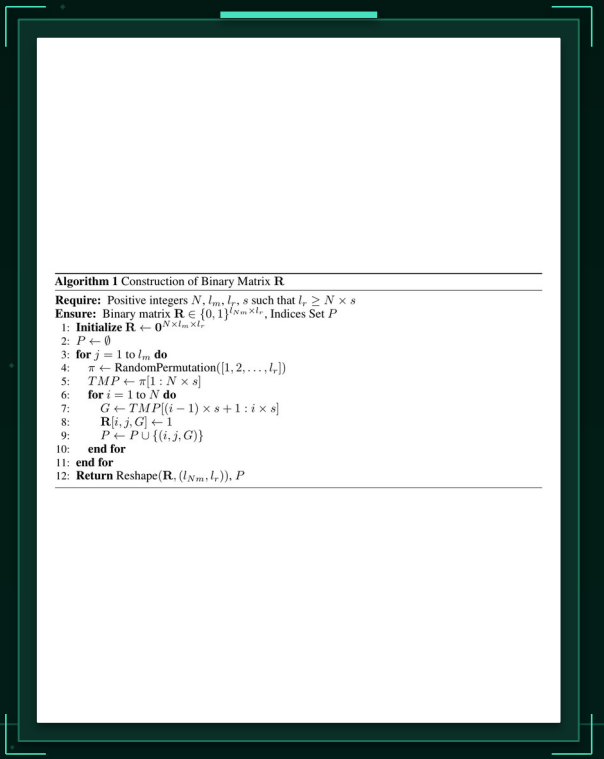
可逆混合

GF(2) 上计算
z(1) = T x

3

恢复依据

逆变换＋多数投票
N 采用奇数



Algorithm 1 Construction of Binary Matrix $\mathbf{R}$

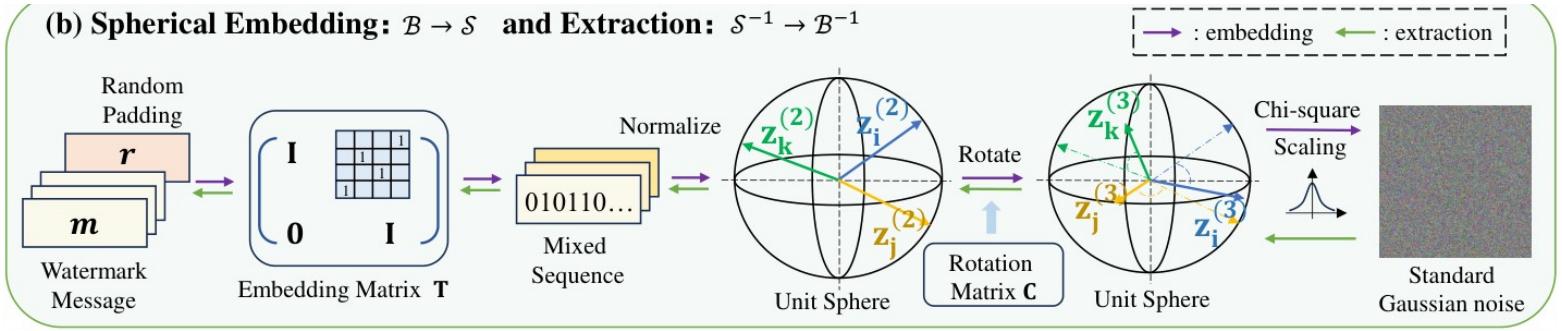
Require: Positive integers N, l_m, l_r, s such that $l_r \geq N \times s$
Ensure: Binary matrix $\mathbf{R} \in \{0,1\}^{l_{Nm} \times l_r}$, Indices Set P

Initialize $\mathbf{R} \leftarrow \mathbf{0}^{N \times l_m \times l_r}$
$P \leftarrow \emptyset$
for $j = 1$ to l_m **do**
  $\pi \leftarrow \text{RandomPermutation}([1, 2, \ldots, l_r])$
  $TMP \leftarrow \pi[1 : N \times s]$
  for $i = 1$ to N **do**
    $G \leftarrow TMP[(i-1) \times s + 1 : i \times s]$
    $\mathbf{R}[i, j, G] \leftarrow 1$
    $P \leftarrow P \cup \{(i, j, G)\}$
  end for
end for
Return $\text{Reshape}(\mathbf{R}, (l_{Nm}, l_r))$, P

恢复冗余与可逆混合共同工作，仍需评估实际反演误差

核心方法二

球面映射与随机半径：构造生成端潜在噪声



$$\mathbf{v} = 2\,\mathbf{z}^{(1)} - \mathbf{1}, \mathbf{z}^{(2)} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \mathbf{z}^{(3)} = \mathbf{C}\,\mathbf{z}^{(2)},$$
$$\text{draw } r \text{ such that } r^2 \sim \chi^2(l_x), \mathbf{z}_w = r\,\mathbf{z}^{(3)}. \tag{10}$$

01 单位球面

二进制信号归一化

02 正交旋转

由签名矩阵 C 变换

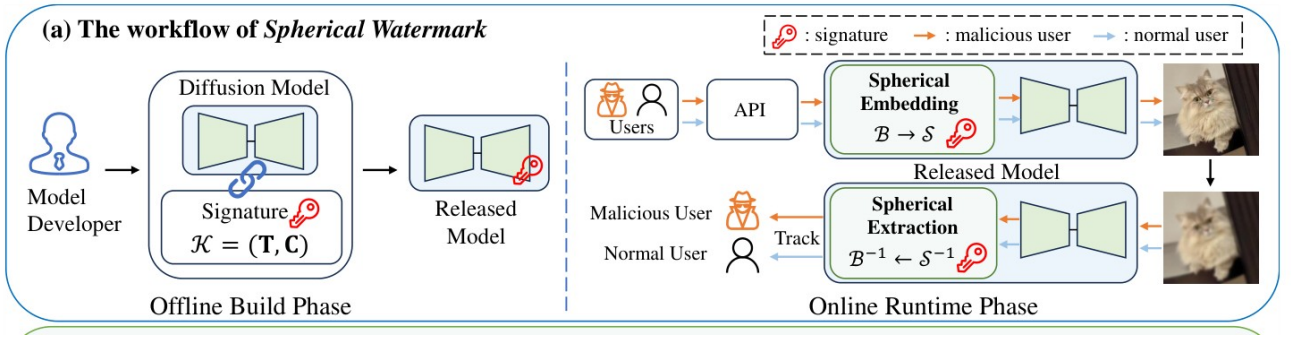
03 随机半径

平方半径服从卡方分布

采用球形水印的生成适配方法，而非在成图上叠加可见纹理

核心方法三

图像反演与逆映射：从收到的图片恢复身份



图像反演

VAE 编码＋扩散反演

逆映射

旋转逆变换＋二进制解混合

消息核验

多数投票＋有效性检查

无法恢复有效标识时，应输出“暂时无法确认”

应用框架

算法层承接论文方法，应用层连接来源记录

01 匿名身份

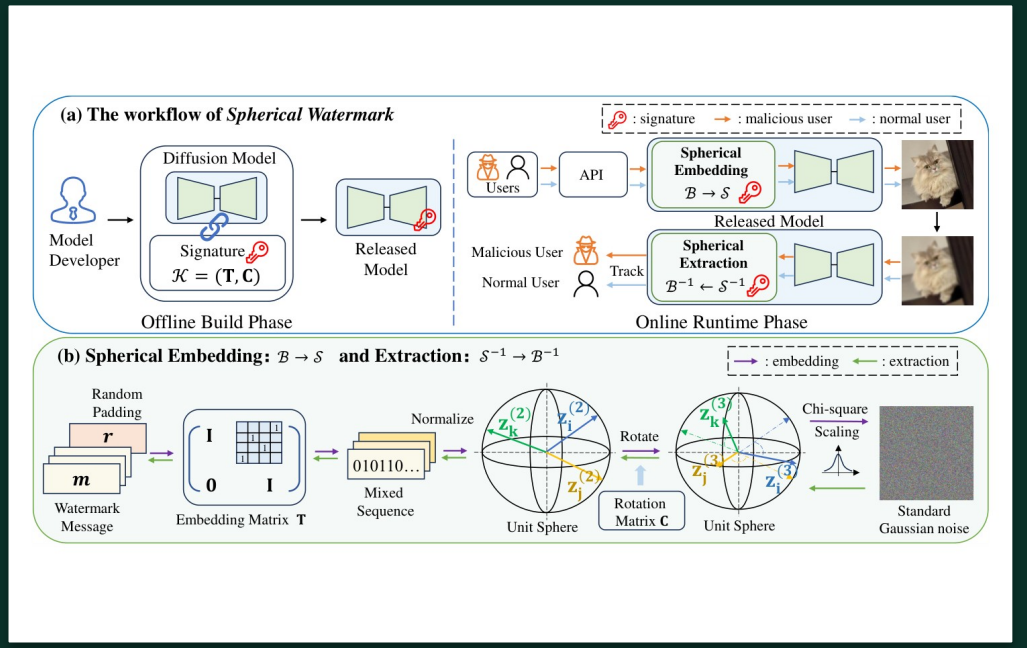
作品编号与署名声明分开存储

02 可信核验

检测成功后再关联记录

03 权限边界

授权材料与核验结果分开展示



- 签名管理
- 素材登记
- 核验记录
- 人工复核

水印是来源线索的载体；登记、授权与权属判断各有职责

实施成果

用可复查的软件证据说明完成度

当前原型

图片上传与登记

后处理水印接口

压缩／缩放／裁剪

来源查询与历史记录

软件验证

13 / 13

现有工作流单元测试通过
包括持久化与错误语义

错误处理与持久化



SPHERICAL WATERMARK: ENCRYPTION-FREE, LOSSLESS WATERMARKING FOR DIFFUSION MODELS

Xiaoxiao Hu[1], **Jiaqi Jin**[1], **Sheng Li**[1], **Wanli Peng**[2], **Xinpeng Zhang**[1], **Zhenxing Qian**[1,*]
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Diffusion models have revolutionized image synthesis but raise concerns around content provenance and authenticity. Digital watermarking offers a means of tracing generated media, yet traditional schemes often introduce distributional shifts and degrade visual quality. Recent lossless methods embed watermark bits directly into the latent Gaussian prior without modifying model weights, but still require per-image key storage or heavy cryptographic overhead. In this paper, we introduce **Spherical Watermark**, an encryption-free and lossless watermarking framework that integrates seamlessly with diffusion architectures. First, our binary embedding module mixes repeated watermark bits with random padding to form a high-entropy code. Second, the spherical mapping module projects this code onto the unit sphere, applies an orthogonal rotation, and scales by a chi-square-distributed radius to recover exact multivariate Gaussian noise. We theoretically prove that the watermarked noise distribution preserves the target prior up to third-order moments, and empirically demonstrate that it is statistically indistinguishable from a standard multivariate normal distribution. Adopting Stable Diffusion, extensive experiments confirm that Spherical Watermark consistently preserves high visual fidelity while simultaneously improving traceability, computational efficiency, and robustness under attacks, thereby outperforming both lossy and lossless approaches.

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从图片保护到传播后核验，以统一接口连接模型与应用

项目实施

围绕方法、产品、验证建立可交付分工

1

方法复现

签名与噪声构造
反演与消息提取

交付：可复现命令与环境

2

产品实现

图片保护与溯源
登记与核验记录

交付：原型与演示录像

3

验证与调研

负样本与传播测试
真实用户任务观察

交付：日志、数据与反馈

以方法、产品与验证三类交付物组织协作

实施条件

论文、代码、算力与登记安全共同支撑落地



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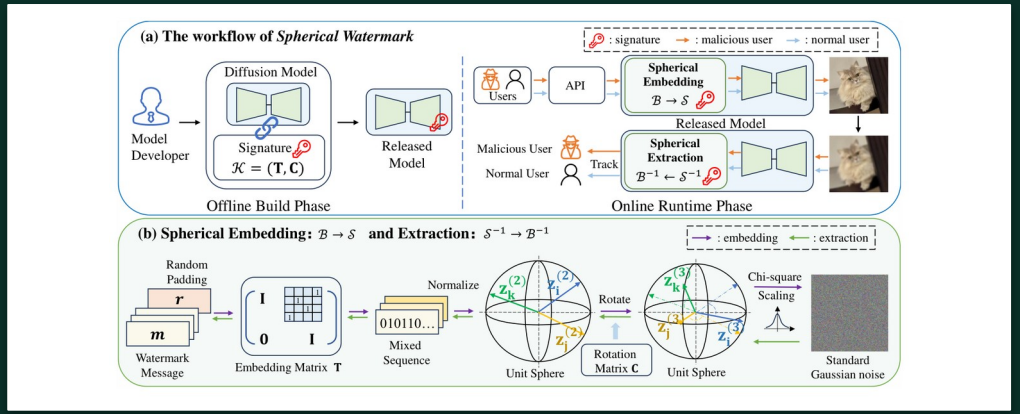
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研究基础

Spherical Watermark · ICLR 2026

对比路线

Tree-Ring / Gaussian Shading / PRC

计算资源

GPU 生成与反演
记录模型与采样配置

数据资源

自有或授权素材
无水印负样本

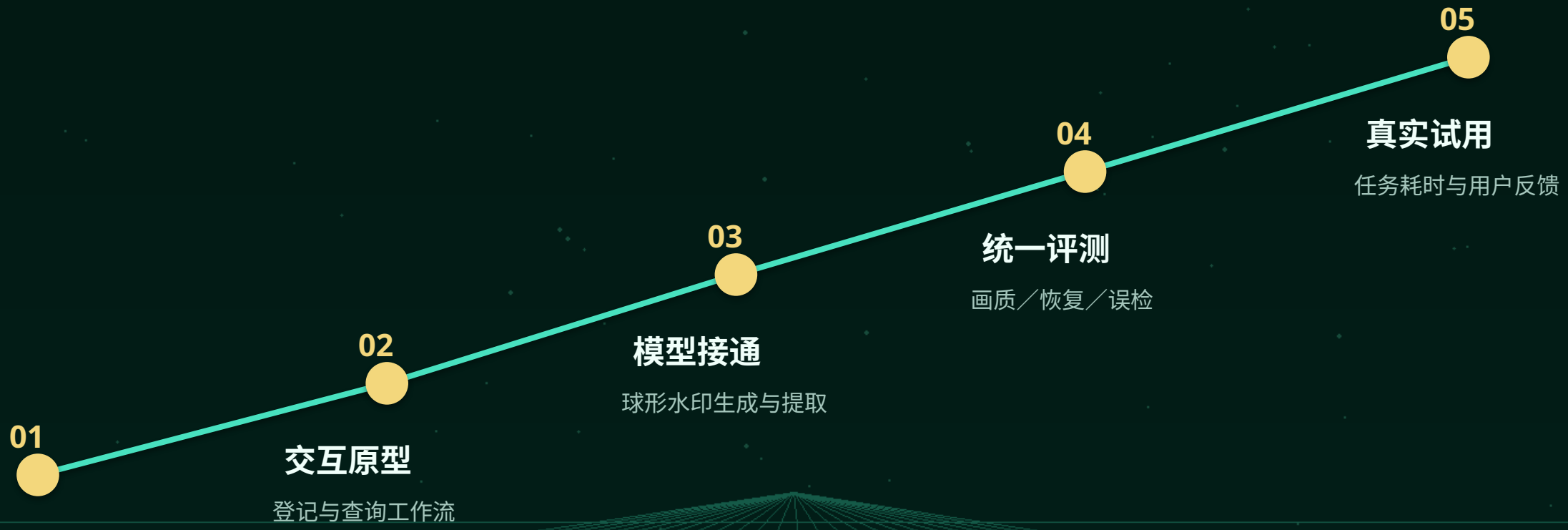
安全边界

受控签名与登记访问
不嵌入个人敏感信息

技术壁垒体现在稳定复现、低误检核验和可审计流程的结合

研发路线

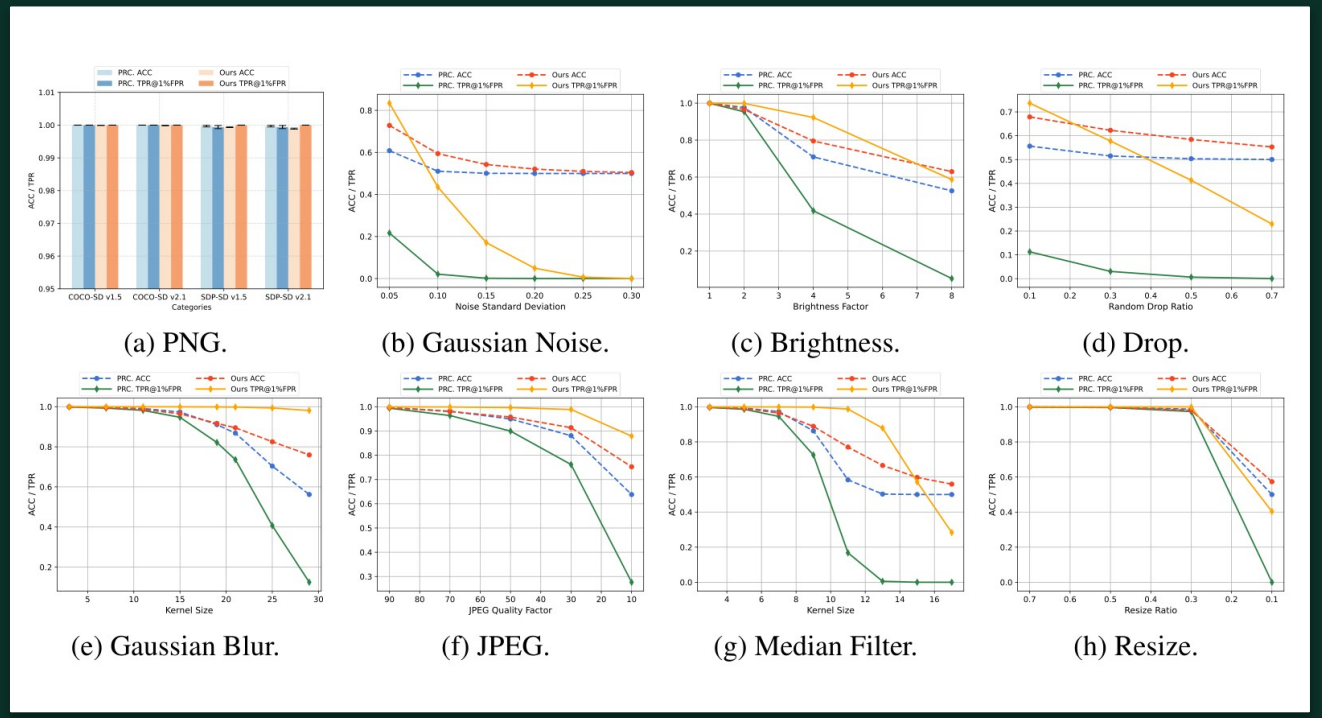
先完成模型链路，再验证应用收益



当前原型已有基础；方法闭环、统一评测与用户任务依次推进

测试与验证

论文证据说明潜力，项目实测检验可用性



(a) PNG. (b) Gaussian Noise. (c) Brightness. (d) Drop.

(e) Gaussian Blur. (f) JPEG. (g) Median Filter. (h) Resize.

01

画质与观感

质量指标＋局部细节

02

传播恢复

压缩／缩放／裁剪

03

误检与混淆

负样本＋错误签名

04

应用体验

正确判断＋任务耗时

同时报告成功、失败与无法确认；不只展示成功样例

方案比较

区分技术能力与应用能力，按同一任务对比

可见署名	文件元数据	后处理水印	球形水印	图溯方案
成图叠字	附带信息	修改已成图	生成噪声嵌入	**身份＋登记关联**
可直接看见	观感不变	适用上传图	生成链路适配	**收到图即可核验**
可能被覆盖	可能被移除	性能依模型	依赖图像反演	**失败进入复核**
需另存声明	需兼容读取	仍需登记关联	论文技术基础	**登记与核验闭环**

在同一来源查找任务下，比较保护能力与使用成本

应用拓展

从一个可验证任务，延伸到多个图像使用场景

首个验证点

电商营销素材
生成—交付—再次核验

相邻场景

创意设计与数字文创
作品流转与来源查找

后续探索

媒体内容与公共传播
系统来源辅助认证

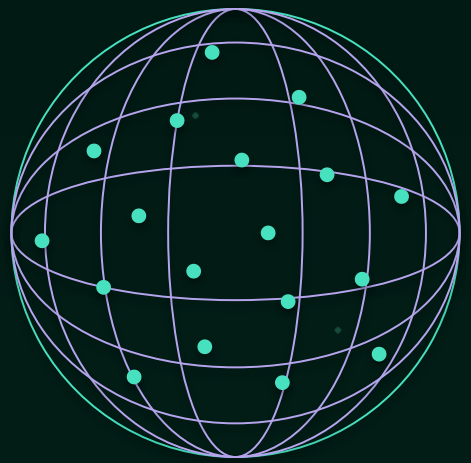
以验证结果推动场景延伸，避免把尚未落地的范围写成成果

社会价值

为图像使用者提供可核实的来源线索

来源认证

辅助核对图片是否匹配
预先登记的生成身份



可信溯源

把恢复的标识关联到
受控登记与历史记录

权益保护

为署名、授权核对与争议
提供可复查的辅助线索

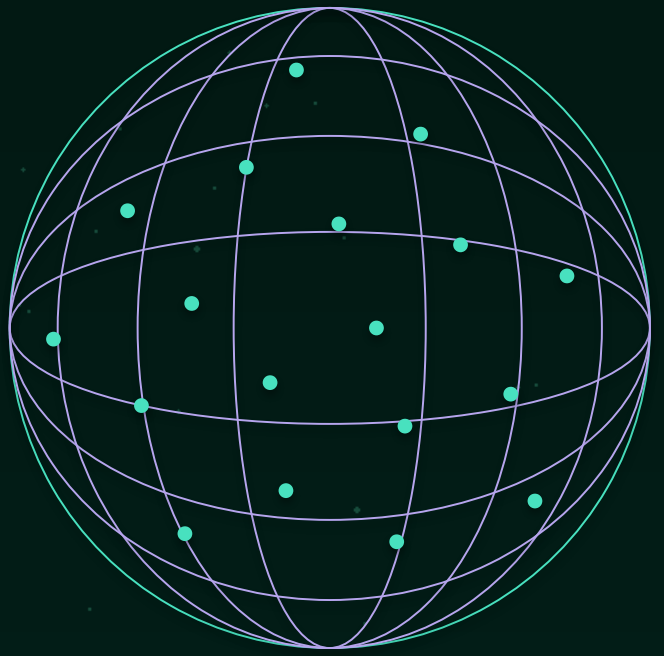
责任边界

不把检出标识等同于
事实真实或法律确权

让身份线索随图传播，让使用决策有据可查

让每一次传播，都保留可核验的身份线索

来源可认证
传播可追溯
权益有线索



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